

Experiment 3: Software Design Principles

Identify stakeholders and perform requirement elicitation for the selected software project using suitable techniques such as interviews, questionnaires, observation, or user personas; prepare user stories and the Requirement Gathering Report.

Project Title: AI Career Guidance & Skill Recommendation System

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Aim

To identify stakeholders and perform requirement elicitation for the AI Career Guidance & Skill Recommendation System using suitable elicitation techniques, prepare user stories, and develop a Requirement Gathering Report.

Objectives

- Identify key stakeholders of the AI Career Guidance & Skill Recommendation System.
- Apply requirement elicitation techniques to understand student and administrator needs.
- Gather functional and non-functional requirements.
- Prepare user stories with acceptance criteria.
- Document the Requirement Gathering Report.

Introduction

Requirement elicitation is the process of discovering stakeholder needs and expectations. For a career guidance platform, requirements focus on extracting student information from resumes, matching profiles with career requirements, identifying skill gaps, creating personalized learning roadmaps, recommending courses/resources, and presenting useful analytics. Common elicitation techniques include interviews, questionnaires, observation, workshops and user personas.

Selected Project

Project Title: AI Career Guidance & Skill Recommendation System

Domain: Education, Career Guidance and Skill Development

Stakeholder Identification

Stakeholder	Role	Requirements
Student / Job Seeker	Uses the platform for career guidance	Easy resume upload, relevant career suggestions, skill-gap analysis and roadmap
Career Counsellor	Reviews career guidance information	Clear profile insights, career requirements and understandable recommendations
System Administrator	Manages career, skill and course data	Manage predefined career requirements, resources and user/system data
Recruiter / Industry Representative	Provides industry-oriented skill expectations	Current role requirements and demanded skills for career-wise comparison
AI Service / Claude API	Extracts and analyzes resume/profile information	Reliable API interaction and accurate structured extraction

Requirement Elicitation Techniques

Technique	Purpose	Outcome
Interview	Discuss career guidance needs with students/counsellors	Detailed functional requirements
Questionnaire	Collect feedback from multiple students	User expectations and preferences
Observation	Observe how students search for careers and learning resources	Usability and workflow requirements
User Persona	Model representative student/job-seeker profiles	User-centred requirements

Sample Requirement Gathering

Interview Findings:

- Students need simple resume uploading and profile extraction.
- Career suggestions should be based on the student's actual skills, education, projects and interests.
- Skill gaps should be clearly shown for the selected career.
- The roadmap should provide an understandable sequence of skills/resources to learn.

Observation Findings:

- Users prefer simple navigation and a clear dashboard.
- Frequently needed information such as career fit, missing skills and roadmap should be easy to access.

Questionnaire Findings:

- Students expect personalized career suggestions rather than generic lists.
- Users value relevant courses/resources linked to specific skill gaps.
- Users want an interactive chatbot for career and preparation-related questions.

User Personas

Persona	Age	Goal	Pain Point
College Student	20	Find a suitable career path and roadmap	Unclear career direction
Final-Year Student	22	Identify skill gaps before placements	Does not know which skills recruiters expect
Career Switcher	26	Move into a new technical role	Difficulty identifying transferable and missing skills

User Stories

ID	User Story	Acceptance Criteria
US1	As a student, I want to upload my resume so that my skills, education, projects and experience can be extracted.	Resume is uploaded successfully and relevant profile information is extracted into structured fields.
US2	As a student, I want suitable career recommendations based on my profile.	System compares the profile with predefined career requirements and displays relevant roles.

User Stories

ID	User Story	Acceptance Criteria
US3	As a student, I want to see my skill gaps for a selected career so that I know what I need to learn.	Existing and missing skills are clearly displayed for the selected career.
US4	As a student, I want a personalized learning roadmap based on my missing skills.	Roadmap provides an ordered set of learning steps/resources relevant to the identified gaps.
US5	As a student, I want course/resource recommendations for each skill gap.	Relevant learning resources are mapped to the required skills.
US6	As an administrator, I want to view analytics about demanded skills and career choices.	Dashboard displays career-wise skill requirements, common gaps, popular choices and demanded skills.
US7	As a student, I want to ask an AI chatbot questions about careers, skills and preparation.	Chatbot responds to career, skill, roadmap and preparation queries.

Requirement Gathering Report

The elicitation process identified resume analysis, profile extraction, career recommendation, skill-gap analysis, personalized roadmap generation, course/resource recommendation, analytics, and an AI career chatbot as the primary functional requirements. Stakeholder needs emphasised personalization, ease of use, clear career information, reliable profile analysis and useful learning guidance.

Functional Requirements:

- Resume upload and AI-based extraction of skills, education, projects and experience.
- Career matching against predefined career requirements such as Java Developer, Full Stack Developer and Data Analyst.
- Skill-gap identification for a selected career.
- Personalized learning roadmap generation.
- Course/resource recommendations for skill gaps.
- Analytics dashboard for demanded skills, popular career choices, common gaps and career-wise requirements.
- AI chatbot for career and preparation queries.

Non-Functional Requirements:

- Usability: simple and student-friendly interface.
- Performance: career/profile information should be presented without unnecessary delay.
- Reliability: resume processing and recommendations should handle valid inputs consistently.
- Security: user resumes and profile information should be handled securely.
- Maintainability: career and skill requirement data should be easy to update.

Technology Stack

Frontend: React **Backend:** Node.js / Express **Database:** MongoDB

AI: Claude API **Analytics:** Python, Pandas

Observation

Using multiple elicitation techniques improves requirement completeness and reduces ambiguity. For this platform, combining student feedback with persona-based analysis helps ensure that the system addresses both career-discovery needs and practical skill-development needs.

Result

Stakeholders were identified, requirements were elicited using suitable techniques, user stories were prepared, and the Requirement Gathering Report for the AI Career Guidance & Skill Recommendation

System was successfully completed.